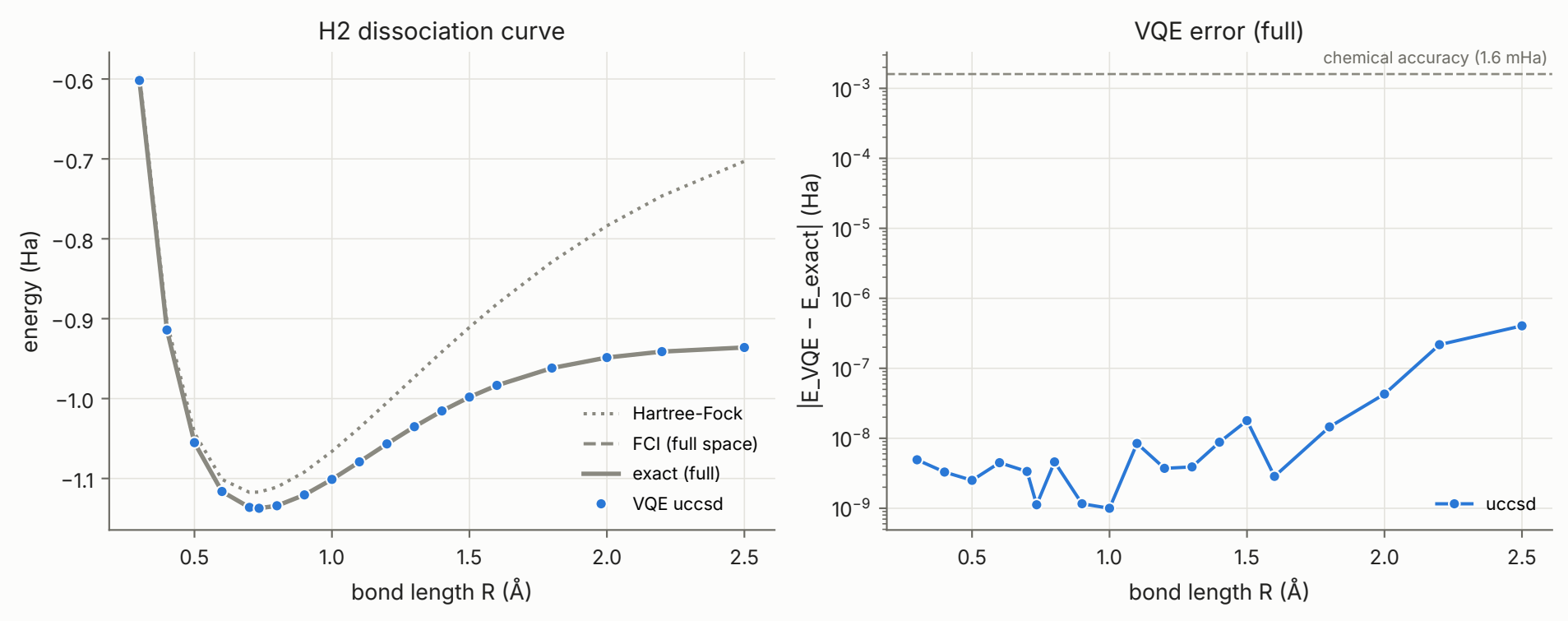
H2 dissociation curve
energy (Ha)
−0.6
−0.7
−0.8
−0.9
−1.0
−1.1
0.5
1.0
1.5
2.0
2.5
bond length R (Å)
Hartree-Fock
FCI (full space)
exact (full)
VQE uccsd
VQE error (full)
chemical accuracy (1.6 mHa)
|E_VQE − E_exact| (Ha)
10^{-3}
10^{-4}
10^{-5}
10^{-6}
10^{-7}
10^{-8}
10^{-9}
0.5
1.0
1.5
2.0
2.5
bond length R (Å)
uccsd